

GENERAL THETA FOUNDATIONS I: ALPHABET DYNAMICS AND CALIBRATED NUMERICAL MEMORY

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ABSTRACT. We distinguish the geometry of a numerical experiment from the dynamics of its command alphabet. For a planar alphabet consisting of the identity and r rotations whose angles satisfy a dual badly approximable inequality, we prove that the minimum hidden stochastic width is $\Theta(N^{r/(2r+1)})$, including every fixed sufficiently small uniform error. Every finite-block optimized action gap is zero. Thus one circle action, one orbit space and one minimum orbit dimension support infinitely many distinct width exponents. The proof pairs a multivariate command-packet converse with an exact common-row polygon realization. For general orthogonal actions we optimize the command law, characterize finite-block expansion by the spectral radius of the uniform action average, and give a rational two-letter example with zero optimized one-step gap but positive two-step gap. We also turn the query-calibrated activation into a finite algebraic optimization, with a semidefinite upper certificate, explicit multiplicity examples and query-frame perturbation bounds. The resource throughout is nonuniform clocked atomic-row label width: tables and arithmetic are free, the horizon is fixed, and one terminal numerical query is answered.

1. INTRODUCTION

A finite numerical experiment prescribes probabilities, rather than only signs of probabilities relative to a threshold. Even when its linear response space has fixed dimension, a positive stochastic realization may need a growing number of persistent labels. Two features of the specification must be kept separate: the compact action generated by its commands, and the finite words that implement that action. The second feature is not determined by the first.

The experiment studied here has response

$$(1.1) \quad \mathbb{P}(B = b \mid x, w, j) = \frac{1 + b\rho\langle e_j, A_w x \rangle}{2}, \quad A_w = A_{a_N} \cdots A_{a_1}.$$

The seed set is finite, the command alphabet is fixed, and exactly one coordinate query arrives after N commands. A simulator must satisfy the specification on every word. Its rows may vary with the epoch, and the simulator may be redesigned for each N . The minimum peak label count at binary total-variation error at most ε is denoted by $W_{N,\varepsilon}$.

For an expanding irreducible action, the earlier occupation and orbit-synthesis theorems give a width exponent equal to half the smallest nonzero orbit dimension. Their proofs are reproduced in the appendices. Expansion is sufficient, but neither a description of all alphabets nor a necessary condition for a growing width. Our first result addresses a class in which every finite-block norm-gap certificate vanishes.

Theorem 1.1 (A nongapped alphabet hierarchy). *Let $r \geq 1$ be fixed. Suppose $\alpha \in \mathbb{R}^r$ and $b_* > 0$ satisfy*

$$(1.2) \quad \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \geq b_* \|m\|_1^{-r} \quad \text{for every } m \in \mathbb{Z}^r \setminus \{0\}.$$

Take the planar commands $I, R_{2\pi\alpha_1}, \dots, R_{2\pi\alpha_r}$, seeds $\{\pm e_1, \pm e_2\}$, and the two coordinate queries. For fixed $0 < \rho \leq 1/10$ and $0 \leq \varepsilon < \rho/(2\sqrt{2})$,

$$(1.3) \quad W_{N,\varepsilon} = \Theta_{\alpha,r,\rho,\varepsilon}(N^{r/(2r+1)}).$$

The lower bound allows arbitrary hidden stochastic states and time-dependent rows. An exact upper realization has common command rows and an N -dependent decoder. The same order holds after adjoining a reflection. Every finite-block optimized action gap of either alphabet is zero.

The hypotheses have explicit examples in every r : put $\xi = 2^{1/(r+1)}$ and $\alpha_i = \xi^i$. An algebraic-norm calculation proves (1.2) with $b_* = (2/9)^r$. The rotation matrices in this example have computable-real entries; algebraicity of the angles is not asserted to make those entries algebraic. The cases $r = 1, 2, 3$ give exponents $1/3, 2/5, 3/7$. The connected effective action is the standard $\text{SO}(2)$ action in all of them. Its minimum orbit dimension is always one. Section 6 proves the theorem, including every command in a packet and every persistent upper-state label.

A second distinction already occurs before the genuinely nongapped regime. Let V_a be the Koopman command operators on the complement of invariant functions. For a probability law p , write

$$g(p) = 1 - \|\sum_a p(a)V_a\|^2, \quad g_1^*(\mathcal{A}) = \max_{p \in \Delta(\mathcal{A})} g(p).$$

The optimized quantity depends only on the alphabet and action. Repeating this optimization over laws on actual length- L words gives g_L^* . We prove that some $g_L^* > 0$ if and only if the uniform one-step average has spectral radius less than one. This need not mean its norm is less than one. A two-letter rational $\text{SO}(3)$ alphabet has $g_1^* = 0$ but $g_2^* > 0$, and consequently linear numerical width. The proof of its positive block gap uses an explicitly identified qualitative algebraic spectral-gap theorem, not a numerical harmonic truncation.

The error tolerance is also part of the actual experiment. The canonical activation constant involves both an irreducible copy and the physical query basis. Section 4 gives an exact finite projection formulation, a semidefinite relaxation with a dual certificate, and a frame reconstruction problem with quantitative stability. It distinguishes these finite problems from infinite-dimensional gap positivity.

Notation	Meaning
$W_{N,\varepsilon}$	Peak available atomic labels; includes a held binary output.
g_L^*, r_L	Optimized length- L squared gap and complementary norm.
s_λ	Smallest orbit dimension in a nontrivial irreducible type.
c_λ	Seed activation divided by the norm of query reconstruction.
b_*, r	Dual Diophantine constant and number of rotation angles.

The new assertions are the multi-angle nongapped hierarchy, the alphabet/block formulation and its worked separation, and the finite calibration and stability results. The single-angle packet idea, entropy occupation estimate, uniform orbit bound and minimum-orbit compiler are inherited ingredients, identified as such. Convex spectral optimization, commutant structure, Diophantine approximation and transformation-group geometry are classical tools. No priority claim for those tools is made. The statement is about label width, not uniform computational space, total branching-program size or an exact finite integer optimum.

2. THE EXPERIMENT AND ITS CHARGED STATE

Fix orthogonal matrices $A_a \in O(\mathbb{R}^D)$ indexed by a finite alphabet $\mathcal{A}$, a nonempty finite set X of unit seeds, $0 < \rho \leq 1/10$, and a horizon $N \geq 1$. The target is (1.1).

Definition 2.1 (Clocked atomic-row realization). A realization consists of finite sets $S_0, \dots, S_N$, initial rows $E_x \in \Delta(S_0)$, stochastic matrices $T_{t,a} : S_t \rightarrow \Delta(S_{t+1})$, and columns $d_j : S_N \rightarrow [-1, 1]$. Its mean output on (x, w, j) is

$$E_x T_{0,a_1} \cdots T_{N-1,a_N} d_j.$$

It has error at most ε if this differs from $\rho\langle e_j, A_w x \rangle$ by at most 2ε for every triple. Its width is $\max(2, |S_0|, \dots, |S_N|)$. Available labels count, including unused padding. The factor two includes a charged held output after the final Bernoulli row.

The seed and query are atomic input symbols. Only the updated label persists: no seed, previous command, persistent random tape, protocol tag or acquisition history is free. A row may sample an arbitrary known real distribution. The epoch, horizon, row table, table construction and lookup, and real arithmetic are free. Common-row constructions below use the same T_a at every command epoch, but their final decoder may depend on N . They are not anytime simulators. A rational fair-bit implementation is a different resource, with sampler states and a self-paced interface; no equality with the atomic optimum is inferred.

Let $H = \langle A_a \rangle$ and let R be the invariant span of HX . Unless specified otherwise, geometric applications assume H connected. Its nontrivial real isotypic subspaces in R are denoted by R_λ , with irreducible models E_λ of dimension d_λ . Orthogonal projections and adjoints refer to the physical Euclidean structure. An isometric intertwiner $J : E_\lambda \rightarrow R_\lambda \subseteq \mathbb{R}^D$ is not an input to the machine.

2.1. Proof-side conditional centroids. A wordwise converse may choose a distribution on commands without changing the specification. If commands are fresh and independent under that test law, a unit orbit process Y_t and the actual register satisfy

$$(2.1) \quad Z_t = \mathbb{E}[Y_t | S_t], \quad q_t = \mathbb{E}\|Z_t\|^2, \quad v_t = \mathbb{E}\|Z_t\|^2, \quad Z_{t+1} = \mathbb{E}[A_a Z_t | S_{t+1}].$$

Indeed, conditional on (S_t, a) the future row uses no other past information. The conditional-expectation projection identity and Jensen's inequality give

$$(2.2) \quad q_{t+1} \leq q_t, \quad \Delta_t := \mathbb{E}\|A_a Z_t - Z_{t+1}\|^2 = v_t - v_{t+1}, \quad \sum_{t < N} \Delta_t \leq 1.$$

The same identities hold at boundaries of independent complete command words; their internal commands need not be independent. All internal updates are first integrated into their endpoint kernel. No identity requiring a fresh independent bit is invoked inside such a word.

Lemma 2.2 (Terminal calibration). *For a fixed unit seed, any realization with error ε satisfies*

$$(2.3) \quad q_N \geq \kappa := \rho/\sqrt{D} - 2\varepsilon.$$

*More generally, for an isometric intertwiner J and a seed with $a = \|J^*x\| > 0$, tracking $Y_t = J^*A_w x/a$ yields*

$$(2.4) \quad q_N \geq \rho a/B(J) - 2\varepsilon, \quad B(J) = \|J^*\|_{\ell_\infty^D \rightarrow \ell_2^{d_\lambda}}.$$

Proof. Let $d = (d_1, \dots, d_D)$. Wordwise coordinate error bounds the norm of the difference between its mean and the target by $2\varepsilon\sqrt{D}$. Thus $\mathbb{E}\langle Y_N, d(S_N) \rangle \geq \rho - 2\varepsilon\sqrt{D}$, whereas its absolute value is at most $\sqrt{D}q_N$. For the second assertion apply J^* before pairing with the normalized constituent orbit. The error is at most $2\varepsilon B(J)$ and $\|J^*d(s)\| \leq B(J)$. This proves (2.4). The pairings are analytical devices; only one query is answered. $\square$

For fixed sizes the optimum error is attained: parameters range over a finite product of compact simplices and intervals, and the objective is a finite maximum of continuous functions. Bounds below therefore concern actual optimal final error, irrespective of cancellation between intermediate errors.

3. CANONICAL ALPHABET AND BLOCK GAPS

Let E be a nontrivial orthogonal representation of the effective group H . On $L^2(\mathbb{S}(E))$ write $V_a f(u) = f(A_a^{-1}u)$ and let Π be Haar averaging. All operators below are restricted to $\mathcal{H} = (I - \Pi)L^2(\mathbb{S}(E))$, which each V_a preserves. The complexification is used for adjoints and spectra. Define

$$(3.1) \quad r_1(\mathcal{A}) = \min_{p \in \Delta(\mathcal{A})} \|P(p)\|, \quad P(p) = \sum_a p(a)V_a, \quad g_1^* = 1 - r_1^2.$$

Equivalent orthogonal actions conjugate these operators. Adding a redundant compact presenting factor does not change them. Repeating a letter does not change the set of admissible averages, although it changes the particular uniform law.

Proposition 3.1 (Optimization, full support and simultaneous laws). *The minimum in (3.1) is attained. The function $g(p) = 1 - \|P(p)\|^2$ is concave and 2-Lipschitz in ℓ_1 . Its supremum on full-support laws equals g_1^* . Every full-support law has positive gap if and only if $g_1^* > 0$. If $m = |\mathcal{A}|$ and u is uniform, then*

$$(3.2) \quad g(u) \geq g_1^*/m.$$

More generally, $q(a) \geq \tau p(a)$ for all letters implies $g(q) \geq \tau g(p)$. For finitely many represented types, individual positive optimized gaps can be certified by one common full-support law.

Proof. Unitary norms give $\|P(p) - P(q)\| \leq \|p - q\|_1$ and $\|P(p)\| \leq 1$. Continuity and compactness give attainment, and the square of a norm is convex. Squaring the preceding inequality gives the stated Lipschitz bound for gaps. Approximation of a maximizing law by its mixtures with the uniform law proves the supremum assertion, not attainment by a full-support optimizer.

If $q = \tau p + (1 - \tau)p'$ and $f \in \mathcal{H}$, convexity of the squared norm yields

$$\|P(q)f\|^2 \leq \tau \|P(p)f\|^2 + (1 - \tau) \|P(p')f\|^2 \leq (1 - \tau g(p)) \|f\|^2.$$

This proves domination. Uniform u dominates p/m for every p . Any full-support law dominates a positive multiple of a maximizing law. Finally a mixture $\sum_i w_i p_i$ dominates $w_i p_i$ for each type's chosen law; a further positive uniform mixture gives full support. Alternatively (3.2) applies to all types at once. Optimizers need not be unique. $\square$

Laziness can be priced precisely: if identity is already an allowed letter, adding it with weight $\eta < 1$ loses at most the factor $1 - \eta$ in this certificate. If it is not a letter, silently inserting idle steps changes the word interface. Nor does zero one-step gap imply zero gap after such a modification.

3.1. A dual problem and the limit of finite truncation. For unit $f \in \mathcal{H}$, let $Q(f)$ be the real Gram matrix

$$Q(f)_{ab} = \operatorname{Re}\langle V_a f, V_b f \rangle, \quad \mathcal{C} = \overline{\operatorname{conv}}\{Q(f) : \|f\| = 1\}.$$

It is a compact convex subset of real positive semidefinite m -by- m matrices with diagonal one.

Proposition 3.2 (Finite-dimensional Gram dual). *One has*

$$(3.3) \quad r_1^2 = \min_{p \in \Delta_m} \max_{Q \in \mathcal{C}} p^T Q p = \max_{Q \in \mathcal{C}} \min_{p \in \Delta_m} p^T Q p.$$

If $r_{1,L}$ is the same norm minimum restricted to spherical harmonics of degrees at most L , then $r_{1,L} \uparrow r_1$. These finite problems are convex semidefinite programs when their blocks are supplied. They give upper bounds on g_1^ , not finite certificates of positive full gap without a tail estimate.*

Proof. For fixed p , the squared norm is $\sup_f p^T Q(f)p$. Taking the closed convex hull does not change the supremum. The expression is convex in p and affine in Q ; the compact convex minimax theorem proves (3.3). This is the usual convex spectral-design mechanism, not a new minimax principle; compare the finite-chain optimization in [11].

The harmonic spaces are invariant under orthogonal substitutions and Haar averaging. Their finite orthogonal sums increase densely. Consequently the truncated norms increase pointwise to the full norm. Take a subsequence of truncated minimizing laws converging to p_* . For every fixed L_0 , continuity on its finite blocks gives $\|P(p_*)\|_{\leq L_0} \leq \lim_L r_{1,L}$. Taking the supremum over L_0 gives $r_1 \leq \lim_L r_{1,L}$; the reverse inequality is immediate. The block norm constraints are

$$\begin{pmatrix} tI & P_\ell(p) \\ P_\ell(p)^* & tI \end{pmatrix} \succeq 0 \quad (\ell \leq L),$$

with invariant subspaces removed. They describe a finite SDP. Convergence alone supplies no effective error bound for the omitted degrees. The set $\mathcal{C}$ is finite-dimensional but is specified by all unit vectors in an infinite-dimensional space; its dual is not an effective positivity algorithm. $\square$

3.2. The block invariant. For $L \geq 1$, give every word $w \in \mathcal{A}^L$ its actual operator V_w and put

$$(3.4) \quad r_L = \min_{\pi \in \Delta(\mathcal{A}^L)} \left\| \sum_w \pi(w) V_w \right\|, \quad g_L^* = 1 - r_L^2.$$

A block is L ordinary inputs. This definition supplies a test law, not a larger free input symbol or an extra runtime register.

Theorem 3.3 (Finite-block expansion). *Let $M = m^{-1} \sum_a V_a$ on $\mathcal{H}$. The following are equivalent:*

- (1) $g_L^* > 0$ for some finite L ;
- (2) $\|M^L\| < 1$ for some finite L ;
- (3) the spectral radius of M is less than one.

The equivalence remains true with M replaced by the average under any fixed full-support law. Also $r_{L+L'} \leq r_L r_{L'}$. For finitely many types with positive finite-block gaps, a common length and a common full-support word law work for all of them.

Proof. The uniform product law assigns m^{-L} to every word and hence dominates $m^{-L}\pi$ for any law π . Proposition 3.1, applied to word operators, gives

$$(3.5) \quad 1 - \|M^L\|^2 \geq m^{-L} g_L^*.$$

A fixed full-support law gives the same argument with the smallest product mass in place of m^{-L} . Thus (1) and (2) are equivalent. The spectral-radius formula $\text{spr}(M) = \lim_n \|M^n\|^{1/n}$ gives (2) $\Leftrightarrow$ (3). Concatenating independently chosen word laws proves submultiplicativity. A common multiple of the finitely many good lengths admits a positive gap for every type by repeating each type's good law; a full-support common word law dominates each one. All invariant complements refer to the original effective action. $\square$

This equivalence identifies a genuine difference between norm and spectral radius of a nonnormal action average. It is not a claim that either can be certified by finitely many unaccompanied harmonic calculations. The next sections make the finite numerical calibration explicit, then apply this distinction to a rational alphabet.

4. FINITE OPTIMIZATION OF THE ERROR CALIBRATION

Fix a nontrivial isotypic subspace R_λ of the reachable space and its orthogonal projection S_λ . All norms in this section use the actual coordinate queries of (1.1), not a freely rotated query

basis. Define

$$(4.1) \quad a_\lambda = \max_{x \in X, J} \|J^* x\|, \quad c_\lambda = \max_{x \in X, J} \frac{\|J^* x\|}{B(J)}, \quad J^* J = I,$$

where $J : E_\lambda \rightarrow R_\lambda$ ranges over isometric intertwiners. This is the canonical isotypic convention: a displayed splitting into equivalent copies is not canonical.

Theorem 4.1 (Projection formulation and algebraic evaluation). *Let $d = d_\lambda$ and let $\mathcal{P}_\lambda$ consist of the real matrices satisfying*

$$(4.2) \quad P = P^T = P^2, \quad \text{tr } P = d, \quad P = S_\lambda P S_\lambda, \quad P A_a = A_a P \quad (a \in \mathcal{A}).$$

Then

$$(4.3) \quad c_\lambda^2 = \max_{x \in X, P \in \mathcal{P}_\lambda} \frac{x^T P x}{\max_{\sigma \in \{-1, 1\}^D} \sigma^T P \sigma}, \quad \frac{a_\lambda}{\sqrt{D}} \leq c_\lambda \leq \frac{a_\lambda}{\sqrt{d}}.$$

If the finite matrices, seeds and the certified isotypic projection are real algebraic, this number and an optimizing projection are real algebraic and can be obtained by a finite quantifier-elimination procedure.

Proof. Norm duality and the extreme points of a cube give

$$(4.4) \quad B(J) = \max_{\|u\|=1} \|Ju\|_1 = \max_{\sigma \in \{-1, 1\}^D} \|J^* \sigma\|.$$

The projection $P = JJ^*$ satisfies (4.2). Conversely any such projection has an invariant image inside one isotypic component. Its real dimension d forces that image to be one irreducible copy, so an isometric intertwiner onto it exists. The value of both numerator and denominator depends only on P , proving the identity. Averaging $\sigma^T P \sigma$ over independent signs gives $\text{tr } P = d$, whereas $\sigma^T P \sigma \leq \|\sigma\|^2 = D$. Taking maxima yields the two inequalities.

The set of projections is compact and nonempty; the denominator is at least d . For any proposed squared value t , feasibility is the finite disjunction over seeds of (4.2) and

$$(4.5) \quad x^T P x \geq t \sigma^T P \sigma \quad \text{for every } \sigma \in \{-1, 1\}^D.$$

These are polynomial conditions of degree at most two in (t, P) . Real algebraic input is represented by defining polynomials and isolating intervals. Quantifier elimination over real closed fields, for example cylindrical algebraic decomposition [8], determines the compact feasible interval of t , its endpoint and a witness. Compact semialgebraic optimization over algebraic data has an algebraic endpoint and an algebraic witness. This is finite decidability, not a polynomial-time claim. There are exponentially many sign constraints. The certified projection is part of the input; no algorithm for discovering a compact-group spectral gap is hidden in this calculation. $\square$

4.1. The multiplicity models and a semidefinite certificate. The real commutant of an isotypic component is a matrix algebra over $\mathbb{F} = \mathbb{R}, \mathbb{C}$ or the quaternions. Its invariant real orthogonal projections of rank d correspond to rank-one Hermitian projections over $\mathbb{F}$. This follows by identifying the isotypic component with a multiplicity space tensored over the irreducible division-algebra commutant: invariant submodules correspond to $\mathbb{F}$ -linear multiplicity subspaces, and adjoints correspond to Hermitian adjoints. In particular the rank constraint is essential in every commutant type. It is not replaced by a trace condition alone.

For a matrix M , write

$$\mathcal{E}_\lambda(M) = \int_H h S_\lambda M S_\lambda h^T dm_H(h).$$

This is the orthogonal projection onto the self-adjoint commutant supported on R_λ . With algebraic command matrices it can equivalently be computed as the orthogonal projection onto the solution

of the finite linear commutation equations. The integral explains its positivity; an integration oracle is unnecessary for that finite linear problem.

Proposition 4.2 (A primal–dual upper certificate). *For each seed x , let v_x be the optimum of*

$$(4.6) \quad \begin{aligned} & \text{maximize } \text{tr}(xx^T Q), \\ & \text{subject to } Q \succeq 0, \quad Q = S_\lambda Q S_\lambda, \quad [Q, A_a] = 0, \quad \sigma^T Q \sigma \leq 1 \quad (\sigma \in \{-1, 1\}^D). \end{aligned}$$

Then $c_\lambda^2 \leq \max_x v_x$. The dual for fixed x is

$$(4.7) \quad \min_{y_\sigma \geq 0} \sum_\sigma y_\sigma, \quad \sum_\sigma y_\sigma \mathcal{E}_\lambda(\sigma \sigma^T) - \mathcal{E}_\lambda(xx^T) \succeq 0 \quad \text{on } R_\lambda.$$

Both optima are attained and equal. An optimal nonzero Q of real rank d certifies equality with the corresponding projection optimization. Higher-rank solutions are only upper bounds.

Proof. For an admissible P , set $Q = P / \max_\sigma \sigma^T P \sigma$. This is feasible and has objective equal to (4.3). The dual follows from the trace pairing in the self-adjoint commutant and positivity of Q . The feasible set is bounded: averaging its sign inequalities gives $\text{tr } Q \leq 1$. A sufficiently small positive multiple of S_λ is strictly feasible relative to the supported commutant. Taking all dual sign weights equal and large makes the dual constraint positive definite on R_λ , because the sign sum is a positive multiple of the identity. Finite-dimensional semidefinite strong duality gives equality and attainment. Finally a positive commuting matrix of rank d has one irreducible image; on that image its self-adjoint restriction is scalar. Hence $Q = tP$. Scaling to a saturated sign constraint recovers the original ratio. This last argument does not apply to higher commutant rank. $\square$

Proposition 4.3 (Computed calibrations). *For a multiplicity-one nontrivial representation filling the physical space of dimension D , with unit seeds, $c = 1/\sqrt{D}$. Thus the standard spherical, traceless complex Hermitian projective, and standard block $\mathbb{R}^3 \oplus \mathbb{R}^2$ examples have respectively*

$$c = 1/\sqrt{D}, \quad c = 1/\sqrt{q^2 - 1}, \quad c_3 = 1/\sqrt{3}, \quad c_2 = 1/\sqrt{2}$$

for unit coordinate seeds in their indicated blocks.

For m equivalent real-type copies in orthogonal physical coordinate blocks $(\mathbb{R}^d)^m$, assume the reachable isotypic space is the whole block space. Writing a seed as $x = (x_1, \dots, x_m)$, one has

$$(4.8) \quad c = \frac{1}{\sqrt{d}} \max_{x \in X, i} \|x_i\|.$$

For $d = 3, m = 2$ and the two seeds $(e_1, e_1)/\sqrt{2}$ and $(e_1, -e_1)/\sqrt{2}$, Euclidean activation is $a = 1$ but query calibration is $c = 1/\sqrt{6}$.

Proof. In multiplicity one the sole copy projection is I , and (4.4) gives $B = \sqrt{D}$. For a coordinate block replace I by its block projection. In the real-type multiplicity model, every isometric intertwiner is, up to an orthogonal change in the model, $Ju = (a_1 u, \dots, a_m u)$ with $\sum_i a_i^2 = 1$. Thus

$$B(J) = \sqrt{d} \sum_i |a_i|, \quad \|J^* x\| = \left\| \sum_i a_i x_i \right\| \leq \sum_i |a_i| \max_i \|x_i\|.$$

Taking one nonzero a_i attains the resulting bound. The two displayed seeds activate the full two-copy space, and each block has norm $1/\sqrt{2}$; a diagonal copy attains $a = 1$. $\square$

The last example admits a short dual certificate as well. For the positive diagonal seed use the sign vector $\sigma = (1, 1, 1, 1, 1, 1)$ and dual weight $1/6$. Twirling under the diagonal $\text{SO}(3)$ action gives $\mathcal{E}(xx^T) = \mathcal{E}(\sigma \sigma^T)/6$. A diagonal-copy projection scaled by $1/6$ is a matching feasible primal point. For the other seed change the signs of the last three coordinates. This certifies the value without a nonconvex numerical search.

4.2. Physical query frames. Let $F : \mathbb{R}^D \rightarrow \mathbb{R}^m$ have rows of Euclidean norm at most one, and replace the coordinate means by $\rho F A_w x$. For a copy J that can be reconstructed from these queries, define

$$(4.9) \quad \beta_F(J) = \min_{LF=J^*} \|L\|_{\ell_\infty^m \rightarrow \ell_2^d}, \quad c_{\lambda,F} = \max_{x,J} \|J^* x\| / \beta_F(J).$$

The minimum is finite exactly when $\ker F \subseteq \ker J^*$. Maxima range only over such copies. If there is none, no positive calibration is inferred. Full column rank is a sufficient condition.

Theorem 4.4 (Frame reconstruction and stability). *For fixed J , (4.9) is the finite second-order cone problem*

$$\min t \quad \text{subject to } LF = J^*, \quad \|L\sigma\| \leq t \quad (\sigma \in \{-1, 1\}^m).$$

Its minimum is attained, and terminal calibration becomes $q_N \geq \rho c_{\lambda,F} - 2\varepsilon$. Duplicating a query does not change β_F or $c_{\lambda,F}$. Adding query rows can only increase the calibrated activation; deleting rows can decrease it or destroy reconstructibility.

For full-column-rank frames F, F' with the same number of rows, set $\delta = \|F' - F\|_{2 \rightarrow 2}$. With the same seed set,

$$(4.10) \quad \frac{c_{\lambda,F}}{1 + \delta\sqrt{m}/\sigma_{\min}(F')} \leq c_{\lambda,F'} \leq (1 + \delta\sqrt{m}/\sigma_{\min}(F))c_{\lambda,F}.$$

If finite seed sets have Hausdorff distance at most h , their calibrated activations for fixed F differ by at most $h \max_j \|F_j\|$.

Proof. The sign formula for the operator norm gives the cone constraints. A bounded norm bounds every column of L , so minimizing sequences have convergent subsequences. Feasibility of $LF = J^*$ is precisely the stated kernel condition. Applying L to the vector of decoder means reproduces the target constituent; its norm is at most β_F and its error at most $2\varepsilon\beta_F$. The proof of Lemma 2.2 then applies.

An added query can be ignored. Conversely, if a row is duplicated, sum the columns of L assigned to its duplicates. The new reconstruction norm cannot exceed the old one, since giving equal signs to those duplicate entries is a restriction of the old cube. These two inequalities prove equality for duplication.

Write $\Delta = F' - F$ and use the Moore–Penrose left inverse F'^+ . If $LF = J^*$, then

$$L' = L - L\Delta F'^+ \quad \text{satisfies} \quad L'F' = J^*.$$

Furthermore,

$$\|\Delta F'^+\|_{\infty \rightarrow \infty} \leq \delta\sqrt{m}/\sigma_{\min}(F'),$$

after bounding each output coordinate by its Euclidean norm. Thus $\beta_{F'}(J) \leq \beta_F(J)(1 + \delta\sqrt{m}/\sigma_{\min}(F'))$. Swap the frames for the reverse comparison, take ratios and maxima, and obtain (4.10). Finally $\|J^*(x - x')\| \leq h$. Since $J^*Ju = u$, $1 \leq \beta_F(J)\|FJu\|_\infty \leq \beta_F(J)\max_j \|F_j\|$ for unit u . This proves the seed bound. No uniform stability is implied as a query frame loses rank. $\square$

These computations concern the sufficient terminal correlation used in a width converse. They do not assert that $\rho c/2$ is the exact critical error of the whole experiment. In particular, the algebraic calibration procedure is not an algorithm for deciding infinite-dimensional expansion.

5. A BLOCK-EXPANDED WIDTH THEOREM AND A RATIONAL EXAMPLE

For connected H , let $s_\lambda = \min_{\|u\|=1} \dim(Hu)$ in each active nontrivial type, and put $\sigma_X = \max_\lambda s_\lambda$. Here activation and reachability use the entire invariant seed span, not an arbitrary displayed decomposition into copies.

Theorem 5.1 (Dominant finite-block expansion). *Suppose a type λ with $s_\lambda = \sigma_X$ has $g_L^* > 0$ for a fixed finite L . If R has dimension D_R , $0 < \rho \leq \min(1/10, 1/(8D_R))$, and $0 \leq \varepsilon < \rho c_\lambda/2$, then*

$$(5.1) \quad W_{N,\varepsilon} = \Theta(N^{\sigma_X/2}).$$

The constants may depend on the fixed action, alphabet, block length, seeds, signal and error. Other active types need not have any block gap. The exact upper construction has common command rows and a horizon-dependent decoder.

Proof. Choose a maximizing seed and copy in (4.1), and choose a positive-gap length- L word law. Feed independent complete words with this law, followed by any fixed remainder of length less than L . The uncompressed constituent orbit stays of norm one. Compressing all intermediate computation gives a common stochastic endpoint row for each starting state and word. Conditional centroids at block boundaries satisfy (2.1)–(2.2); deterministic remaining commands cannot increase their mean norm. Terminal calibration therefore gives $q_t \geq \kappa = \rho c_\lambda - 2\varepsilon > 0$ at every block boundary.

The uniform orbit estimate of Theorem A.1 and the occupation theorem in Appendix B, now in dimension d_λ , imply, at peak width K ,

$$(5.2) \quad \lfloor N/L \rfloor g_L^* \leq B_{d_\lambda, A_\lambda, s_\lambda} \kappa^{-3-2/s_\lambda} K^{2/s_\lambda}.$$

Sphere and Lebesgue action norms agree by polar coordinates: both averages act trivially on the radial factor and identically on L^2 of the sphere. Thus no full presenting-group gap is substituted for the defined action gap. This proves the lower order.

For completeness the exact upper bound prices all branches. Split R orthogonally into irreducible copies of dimensions d_i , including trivial copies. Choose branch i initially with probability $w_i = d_i/D_R$ and store its identifier in a disjoint label set. In that branch synthesize the mean $(\rho/w_i)A_w\pi_i x$. The component seed has norm at most one, and $\rho/w_i \leq 1/(8d_i)$. The construction in Theorem C.2, whose initialization also accepts vectors in the unit ball, gives $O(N^{s_i/2})$ labels. A trivial copy needs a fixed two-label mixture to retain its initial scalar; its commands can be identities. The weighted sum of all branch outputs is exactly $\rho A_w x$. Total width is the sum of the branch widths, hence $O(N^{\sigma_X/2})$. Only one final query is answered, and its row is selected from the branch carried by the label. This proves the matching upper bound and prices the selector. $\square$

The full group norm-gap sufficient theorem and the earlier one-step dominant-action theorem are contained in this statement, not withdrawn. Its hypothesis is still sufficient rather than necessary. Section 6 treats a family outside every finite-block expansion certificate.

Theorem 5.2 (A rational alphabet whose first useful block has length two). *Let*

$$a = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/5 & -4/5 \\ 0 & 4/5 & 3/5 \end{pmatrix}, \quad b = \begin{pmatrix} 3/5 & -4/5 & 0 \\ 4/5 & 3/5 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathcal{A} = \{a, ab\}.$$

For the standard $\text{SO}(3)$ action on $\mathbb{R}^3$,

$$g_1^*(\mathcal{A}) = 0, \quad g_2^*(\mathcal{A}) > 0.$$

The second inequality uses the algebraic spectral-gap theorem [2, Theorem 1.2]; it is a qualitative external certificate. With unit coordinate seeds, $0 < \rho \leq 1/24$ and $\varepsilon < \rho/(2\sqrt{3})$, the numerical width is $\Theta(N)$.

Proof. The common angle is irrational relative to 2π . Otherwise $2\cos\theta = 6/5$ would be a rational algebraic integer, hence an integer. Powers of a and b therefore contain the full rotations about their respective axes in their closures. Their infinitesimal rotations generate $\mathfrak{so}(3)$, so the closure is $\text{SO}(3)$. The given two-letter alphabet generates the same closed group since $a^{-1}(ab) = b$.

For any probability p , its average is $V_a((1-p)I + pV_b)$. There is a nonconstant mean-zero function fixed by V_b , for example the third coordinate on the sphere. This average preserves its norm. Every average is a contraction, so every one-step norm is exactly one.

Put $c = a^{-1}ba$. It rotates about an axis different from that of b , and b, c again generate a dense subgroup of $\text{SO}(3)$. They have rational matrices. For the symmetric average

$$Q = (V_b + V_{b^{-1}} + V_c + V_{c^{-1}})/4$$

the cited theorem, applied to the compact connected simple group $\text{SO}(3)$, gives a spectral gap. In particular there is $\delta > 0$ with $\langle (I - Q)f, f \rangle \geq \delta \|f\|^2$ on the mean-zero spherical subspace. The spherical action is a quotient of the regular group action; all entries of the adjoint matrices are algebraic. This is the sole deep input.

The uniform two-step average equals

$$V_{a^2} B, \quad B = (I + V_c)(I + V_b)/4.$$

For its four unitary summands $U_1 = I, U_2 = V_b, U_3 = V_c, U_4 = V_{cb}$, the variance identity gives

$$\begin{aligned} \|f\|^2 - \|Bf\|^2 &= \frac{1}{16} \sum_{i < j} \|U_i f - U_j f\|^2 \\ &\geq \frac{\|f - V_b f\|^2 + \|f - V_c f\|^2}{16} = \frac{1}{4} \langle (I - Q)f, f \rangle. \end{aligned}$$

Thus $g_2^* \geq \delta/4 > 0$. The minimum orbit dimension is two and Proposition 4.3 gives $c = 1/\sqrt{3}$. Apply Theorem 5.1. Finite matrix checks establish the displayed identities, not the external infinite-dimensional value of δ . $\square$

6. SHARP WIDTHS WITHOUT ANY FINITE-BLOCK GAP

We prove Theorem 1.1. Throughout this section $r, \alpha, b_*, \rho, \varepsilon$ are fixed as in that theorem. Distances on $\mathbb{R}/\mathbb{Z}$ use the shorter circular distance in $[0, 1/2]$, and logarithms are natural. We retain the entire wordwise specification; command laws appear only in the converse.

6.1. A packet can only lose conditional amplitude.

Lemma 6.1 (Endpoint quantization of a fresh packet). *Let a complete random command word W be independent of the past and of the old register's private randomness. Its net orthogonal action is U_W . If the register has at most k labels at its endpoint, and the product distribution obeys*

$$(6.1) \quad \sup_{\|u\|=1, v_1, \dots, v_k \in \mathbb{S}^1} \mathbb{E} \max_j \langle v_j, U_W u \rangle \leq 1 - \gamma,$$

then the conditional-mean norms at the two boundaries satisfy $q' \leq (1 - \gamma)q$.

Proof. Let $z_s = \mathbb{E}[Y \mid S = s]$ before the word. Integrate all internal rows and randomness into a stochastic endpoint kernel $T(j \mid s, W)$. Independence of the complete word and the update rule give

$$z'_j = \mathbb{E}[U_W z_s \mid S' = j].$$

Choose $v_j = z'_j / \|z'_j\|$ when $z'_j \neq 0$, and any unit direction otherwise. Then

$$\begin{aligned} q' &= \sum_s \mathbb{P}(S = s) \mathbb{E}_W \sum_j T(j \mid s, W) \langle v_j, U_W z_s \rangle \\ &\leq \sum_s \mathbb{P}(S = s) \|z_s\| \sup_{\|u\|=1} \mathbb{E}_W \max_j \langle v_j, U_W u \rangle \leq (1 - \gamma)q. \end{aligned}$$

No condition is imposed on the number of intermediate states. The maximizing assignment in the second line need not be executable; allowing it only increases an upper bound on surviving amplitude. Commands inside W may be correlated. $\square$

For an integer $k \geq 1$, put

$$(6.2) \quad L = \lceil (2k)^{1/r} \rceil, \quad B = r(L-1), \quad s = b_* B^{-r}.$$

A packet has r segments, each of length $L-1$. In segment i first give J_i copies of rotation i , then $L-1-J_i$ idle letters, where the J_i are independent uniform variables in $\{0, \dots, L-1\}$. Thus there are exactly B actual commands and $L^r \geq 2k$ equiprobable products.

Lemma 6.2 (Separated multivariate packets). *The products in this packet have pairwise circular angular distance at least s . Condition (6.1) holds with $\gamma = s^2$.*

Proof. A difference of distinct count vectors is nonzero with ℓ_1 norm at most B . Inequality (1.2) therefore gives the separation. A circular open cap of radius $s/2$ around one decoder direction contains at most one product point, even after an arbitrary common rotation by u . At most k of the L^r points lie in the union of the k caps. At least half the points have distance at least $s/2$ from every decoder direction. For $0 \leq d \leq 1/2$, $1 - \cos(2\pi d) = 2 \sin^2(\pi d) \geq 8d^2$. On that half the scalar-product loss is at least $2s^2$. Its average is at least s^2 , as claimed. Boundary points satisfy the same weak inequality. $\square$

Proposition 6.3 (Finite packet and occupation bounds). *Let $\kappa = \rho/\sqrt{2} - 2\varepsilon > 0$. For a simulator of peak width K , define B by (6.2) with $k = K$. Then*

$$(6.3) \quad \lfloor N/B \rfloor b_*^2 B^{-2r} \leq \log(1/\kappa), \quad N < B + b_*^{-2} B^{2r+1} \log(1/\kappa).$$

Consequently $K \geq cN^{r/(2r+1)}$ for a fixed positive c . More generally, if B is defined from a proposed small width k , any available profile satisfies

$$(6.4) \quad \#\{1 \leq t \leq N : K_t \leq k\} \leq B - 1 + b_*^{-2} B^{2r+1} \log(1/\kappa),$$

with no bound on other widths.

Proof. Feed independent packets up to time $B \lfloor N/B \rfloor$ and idle letters thereafter. The same realization must pass this law because its requirement is wordwise. Lemmas 6.1 and 6.2 give $q_N \leq (1-s^2)^{\lfloor N/B \rfloor}$. Terminal calibration gives $q_N \geq \kappa$. Taking logarithms and using $-\log(1-s^2) \geq s^2$ proves both inequalities. Since $B = r(\lceil (2K)^{1/r} \rceil - 1) \leq r(2K)^{1/r}$, the second inequality gives the stated lower power, after enlarging constants for finitely many small N .

For the profile statement discard small-width endpoints before B . Among the rest, greedily select the earliest available endpoint and then the next at least B later. At most B selected-or-discarded integer endpoints are accounted for by each choice. The chosen length- B intervals are disjoint. Put an independent packet in each and deterministic idle commands in all gaps. Only the chosen endpoints need have at most k labels. Repeating the loss argument bounds the number of chosen intervals by $s^{-2} \log(1/\kappa)$. Adding the at most $B-1$ initial endpoints proves (6.4). The test law is selected from available layer sizes, not by inspecting a machine's private state. $\square$

6.2. Simultaneous denominators at every scale. The following elementary argument records the arithmetic needed for an upper construction at every horizon, not merely on a subsequence.

Lemma 6.4 (Comparable simultaneous denominators). *For each integer $Q \geq 2$, there are integers $p_1, \dots, p_r, q$ such that*

$$(6.5) \quad c_* Q^r \leq q \leq Q^r, \quad \max_i |q\alpha_i - p_i| \leq Q^{-1}, \quad c_* = (b_*/r^{r+1})^r.$$

Proof. Place the $Q^r + 1$ vectors $j\alpha \bmod \mathbb{Z}^r$, $0 \leq j \leq Q^r$, in Q^r equal boxes. Two lie in the same box; subtraction gives $1 \leq q \leq Q^r$ and the stated errors. This is the usual simultaneous Dirichlet pigeonhole argument.

Put $M = \lfloor q^{1/r} \rfloor$. The $(M+1)^r > q$ integer vectors in $\{0, \dots, M\}^r$ cannot have distinct values of $m \cdot p \bmod q$. Their difference gives a nonzero m with $H = \|m\|_1 \leq rM$ and $q \mid m \cdot p$. Hence

$$b_* H^{-r} \leq \|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} \leq \frac{H}{qQ}.$$

It follows that $b_* qQ \leq H^{r+1} \leq r^{r+1} q^{(r+1)/r}$, which rearranges to the lower bound on q . No unexplained transference constant is used. $\square$

Proposition 6.5 (Exact common-row upper realization). *For every $N \geq 1$ there is an exact realization using $O(N^{r/(2r+1)})$ labels, with common command rows and an N -dependent decoder. It uses the original $r+1$ letters; a reflection may also be included at the same cost.*

Proof. Choose an integer Q so that $c_* Q^r \geq 4$ and

$$\frac{4\pi^2 N}{c_*^2 Q^{2r+1}} \leq \log 2.$$

It can be chosen $O(N^{1/(2r+1)})$, with a constant depending only on the fixed data. Take q, p_i from Lemma 6.4. Let P be the regular q -gon with vertices $u_j = (\cos(2\pi j/q), \sin(2\pi j/q))$. Write $\delta_i = 2\pi(\alpha_i - p_i/q)$, so $|\delta_i| \leq 2\pi/(qQ) \leq \pi/q$. The rotated vertices lie in $\Lambda_i P$, where

$$(6.6) \quad \Lambda_i = \frac{\cos(\pi/q - |\delta_i|)}{\cos(\pi/q)}, \quad \log \Lambda_i \leq |\delta_i| \tan(\pi/q) \leq \frac{4\pi^2}{q^2 Q}.$$

The first identity follows by intersecting the ray to a rotated vertex with the chord between the adjacent polygon vertices. The logarithmic inequality follows by integrating the derivative of $\log \cos(\pi/q - t)$, and $\tan(\pi/q) \leq 2\pi/q$ for $q \geq 4$.

Put $\Lambda = \max(1, \Lambda_1, \dots, \Lambda_r)$. Then $\Lambda^N \leq 2$, and $\Lambda^{-1} R_{2\pi\alpha_i} P \subseteq P$. The same containment holds for the identity and for reflection in the first coordinate axis, since the latter preserves P . For each label u_j and command choose convex coordinates in P of its image divided by Λ as the stochastic row. These choices are made once and do not depend on the epoch. Initially represent $x/2$ as a mixture of the vertices; this is possible since P contains the radius- $\cos(\pi/q)$ disk. At epoch t interpret label j as $2\Lambda^t u_j$. Induction using the common rows gives mean $A_w x$. The final decoder is

$$d_\ell(j) = 2\rho \Lambda^N \langle e_\ell, u_j \rangle,$$

whose modulus is at most $4\rho \leq 2/5$. Every conditional response is therefore exact. At most three successors per row suffice by planar Carathéodory; the vector interpretation is a description of fixed rows, not an extra real register. Finally $q \leq Q^r = O(N^{r/(2r+1)})$. Every partial command step uses the same q labels. $\square$

6.3. Vanishing gaps and explicit angles.

Proposition 6.6 (All block gaps vanish). *For any finite family of planar rotations generating a dense subgroup of $\text{SO}(2)$, all optimized finite-block gaps on its circle action are zero. This remains true after adjoining the reflection above when the effective group is $O(2)$.*

Proof. Simultaneous Dirichlet approximation gives integers $n_j \rightarrow \infty$ for which all $n_j \alpha_i$ approach integers. For an irrational generating family the denominators cannot remain bounded. On the mean-zero Fourier mode $e^{in_j \theta}$ every fixed rotation word then has eigenvalue approaching one. Every average on any fixed finite word set has norm one, so no law has positive gap. For the reflected alphabet use the normalized real function $\cos(n_j \theta)$. Reflection fixes it; rotations and any fixed words move it by a norm tending to zero. The same conclusion follows. The constants are invariant functions only at frequency zero because the effective actions are respectively $\text{SO}(2)$ and $O(2)$. $\square$

Proposition 6.7 (An explicit family in every alphabet size). *Let $\xi = 2^{1/(r+1)}$ and $\alpha_i = \xi^i$, $1 \leq i \leq r$. Then (1.2) holds with $b_* = (2/9)^r$.*

Proof. For nonzero m let $H = \|m\|_1 \geq 1$ and choose $m_0 \in \mathbb{Z}$ nearest to $-\sum_i m_i \xi^i$. The polynomial $P(T) = m_0 + \sum_i m_i T^i$ is nonzero of degree at most r . Since $T^{r+1} - 2$ is Eisenstein, $P(\xi) \neq 0$. It is an algebraic integer of nonzero integer norm. All conjugates of ξ have the same modulus and $|\xi^i| < 2$, so $|m_0| \leq 2H + 1/2 \leq 5H/2$. Each of the other r conjugates of $P(\xi)$ has modulus at most $9H/2$. The product of all $r+1$ moduli is at least one. Therefore

$$\|m \cdot \alpha\|_{\mathbb{R}/\mathbb{Z}} = |P(\xi)| \geq (2/9)^r H^{-r}.$$

The norm argument is classical algebraic Diophantine separation; its role here is to supply explicit input data for the memory theorem. $\square$

Proof of Theorem 1.1. Propositions 6.3 and 6.5 give the matching powers. Adding reflection retains the upper construction and the converse applies to its rotation-only subalphabet. Proposition 6.6 proves the gap assertion. The effective groups are the full indicated circle and orthogonal circle actions, since (1.2) makes each angle irrational. Proposition 6.7 supplies the announced examples. $\square$

At any one internal cut the exact separate positive minimum is still three. Indeed all actual residuals depend affinely on a planar vector, and the idle-prefix images of the four seeds give affine dimension two. Hence at least three normalized states are required. For the reverse bound choose an equilateral triangle of circumradius $2\rho \leq 1/5$. It contains the radius- ρ disk. Its three vertices define legal complete future continuations, since every future command is orthogonal and every final coordinate remains bounded by one. Their convex coordinates factor every actual residual at that cut. One can embed this factorization in a complete machine by retaining the full past before that cut and the chosen vertex plus the full remaining word thereafter. Other cuts are unrestricted in this separate-minimum assertion. Thus all exponents above measure causal compatibility beyond the same static value three.

7. FINITE COMPONENTS AND OBSERVABLE QUOTIENTS

Connectedness is not harmless bookkeeping. If H is a compact matrix group, its identity component H^0 has finite index, and other components can permute its isotypic subspaces. An uncharged component label would therefore change the realization problem.

Proposition 7.1 (A paid finite-component compiler). *Choose representatives r_c of the finite quotient H/H^0 , with $r_e = I$, and write*

$$(7.1) \quad A_a r_c = r_{c'} h_{a,c}, \quad h_{a,c} \in H^0.$$

Suppose the finite connected-component alphabet $\{h_{a,c}\}$ has an exact common-row vector realization of width K_N for the required seeds and signal, whose terminal represented vectors have norm at most one. Then the original alphabet has an exact realization of width at most $\max(2, [H : H^0] K_N)$. The terminal decoder may depend on N . If H itself is finite, the exact width is at most $\max(2, |HX|)$ at every horizon.

Proof. Use labels (c, s) . Given letter a and label (c, s) , update c to the uniquely determined c' in (7.1), and use the connected compiler's row for $h_{a,c}$ on s . The row depends only on the actual stored pair and the input. Inductively, if $A_w = r_c h_w$, the represented internal vector has mean $\rho h_w x$. At the final query use $\langle e_j, r_c v_s \rangle$. Its modulus is at most one and its overall mean is $\rho \langle e_j, A_w x \rangle$. The component identity is part of every label. No fixed choice of H^0 isotypic coordinates is assumed preserved by all components.

For finite H , retain the actual point in the finite orbit union HX , update it by the command permutation, and sample the prescribed bounded terminal row. This gives the stated bound. It is an upper bound, not a claim that every orbit point is a necessary label. $\square$

The minimum-orbit construction in Appendix C, followed by the paid branch construction in Theorem 5.1, supplies such vector compilers at the stated small signal even without expansion. A matching lower bound for an arbitrary finite extension requires information about actual return words and the action on the activated subspaces; it does not follow solely from the index. The reflection extension in Theorem 1.1 is a matched nongapped example: its upper polygon is reflection-invariant, while restriction to rotation words gives the full lower order.

Proposition 7.2 (Metric orbit quotients do not specify numerical memory). *The metric quotients of the standard actions $\mathrm{SO}(d) \curvearrowright \mathbb{R}^d$, $d \geq 2$, are all the half-line. Their minimum nonzero orbit dimensions are $d - 1$. The same half-line is the metric quotient of the finite reflection action $\{\pm 1\} \curvearrowright \mathbb{R}$. Moreover there are numerical alphabets with the identical effective standard $\mathrm{SO}(2)$ action, identical seeds and queries, and identical orbit geometry, but infinitely many distinct sharp width exponents.*

Proof. Two standard orthogonal orbits are spheres of radii r, s ; their distance is $|r - s|$. The reflection quotient has the same distance between the absolute values. Orbit dimensions in the standard connected actions are those of the spheres. Thus metric quotient-equivalence alone cannot preserve the minimum orbit dimension. The last assertion is Theorem 1.1 with the explicit angle families of Proposition 6.7: all act on the same plane and their exponents $r/(2r + 1)$ are distinct. $\square$

An ambient linear isometry taking every orbit onto an orbit does preserve the dimensions of those orbit submanifolds. To preserve the numerical experiment it must additionally intertwine the actual command letters and transform the seeds and queries. An isometry of the quotient, or a polar section with its finite Weyl group, need not do this. These are distinct equivalence relations, not competing proofs of the same invariant.

8. CLASSICAL BOUNDARIES AND SCOPE

8.1. Transformation groups. Gorodski–Lytchak [12, Section 1 and Proposition 1.1] distinguish metric quotient-equivalence from ambient orbit-equivalence and study reductions, boundary, cohomogeneity and copolarity. Their Proposition 1.1 imposes a no-boundary hypothesis when recovering the ambient dimension. The half-line example in Proposition 7.2 has boundary and makes no contrary assertion. Their principal-isotropy reductions and polar examples identify quotient metric spaces; they do not provide command intertwiners preserving specified numerical words. The full alphabet is additional dynamic data.

The sphere-orbifold classification in [13, Theorem 1.1] concerns the geometry of the quotient, not the wordwise positive width of a chosen generating set. Gorodski–Thorbergsson [14, Theorems 1.1 and 1.2] classify variationally complete and irreducible taut representations through orbit equivalence and osculating-space structure. Those results can identify useful orbit types in their classes; neither tautness nor polarity supplies a gap for a given alphabet. Our uniform orbital small-ball estimate is an inherited compact differential-chart argument. It is not presented as a replacement for the slice theorem or a new classification of compact representations. The algebraic evaluation of selected adjoint and exterior-power minima in the supporting manuscript remains intact; no general highest-weight classification is asserted here.

The support-net compiler uses the extrinsic convex hull of an orbit and quadratic loss at a support maximum. These are standard convex and Riemannian geometric facts [7, 10, 9]. Its role is to make one legal set of transition rows serve every layer. The nongapped hierarchy is more

sharply alphabet-dependent: even preserving the full orbit foliation in the same ambient plane does not preserve its width exponent.

8.2. Spectral design, circle walks and branching programs. Boyd–Diaconis–Xiao [11, Sections 2–4] formulate finite reversible Markov-chain design as convex spectral-norm minimization with semidefinite duality. Proposition 3.2 uses that classical convex structure, but its admissible operators act on an infinite spherical harmonic sum. Its truncated programs have the opposite certification direction from an upper tail estimate. We do not infer a positive infinite gap from a finite SDP.

The algebraic expansion theorem of Benoist–de Saxcé [2, Theorem 1.2 and Definition 2.1] is an external qualitative input. In Theorem 5.2 it is used only for the symmetric average of the explicitly verified dense generators b, c . Our four-word variance estimate then supplies the stated two-step norm loss. This separation avoids treating the norm of a nonnormal one-step average as its spectral radius.

Berkes–Borda [15] analyze circle random walks using Fourier multipliers and Diophantine data. The simultaneous pigeonhole argument and the algebraic-norm separation in Section 6 are elementary arithmetic tools. The additional assertion here concerns loss forced by compression into an arbitrary hidden register and a matching exact positive realization. Its test packets are independent as complete words, not independent-letter samples inside a packet. The $r = 1$ mechanism and the general conditional-centroid packet inequality are inherited from the earlier rotation development; the multivariate separation, every-scale denominator argument and their matched $r/(2r + 1)$ law are the extension proved here.

The clocked model is a nonuniform probabilistic ordered read-once layered program with real terminal columns. Regular and permutation branching-program Fourier bounds, such as [5, 6, 3], concern their stated transition assumptions and pseudorandomness or Fourier growth. This paper does not claim those objectives or count total program size. A Bernoulli row converts one bounded column into randomized Boolean acceptance, but preserving a cutpoint sign is weaker than preserving its numerical value. The constant-width attenuated sign simulator, the version-pinned quantum-automata comparison and the projective examples remain in the unaltered supporting manuscript; they are not new claims of this revision.

8.3. Resource ledger in the theorem. For a common-row upper realization with K labels there are $|\mathcal{A}|K$ command rows, in addition to the seed and decoder rows. A bound of $d + 1$ successors per row does not bound the complexity of constructing their weights or storing their real values. For a general clocked machine there can instead be $|\mathcal{A}| \sum_{t < N} K_t$ distinct command rows. None of these descriptions is charged as persistent label width.

The largest toral widths in Theorem 1.1 correspond to $\frac{r}{2r+1} \log_2 N + O(1)$ label bits, not polynomially many bits. Exact irrational stochastic rows are not finite random-bit algorithms. Output error, calibration, alphabet size and representation dimension are fixed before N grows. A simultaneous channel of all coordinate answers, repeated queries, an anytime decoder and adaptive experimental design are different tasks.

8.4. Conclusions. The alphabet can be expanding only after a finite word block, or nongapped at every finite block while still forcing a sharp growing width. Optimizing the proof law makes this distinction canonical. The explicit nongapped hierarchy shows why a representation exponent cannot classify all alphabets, and supplies matching laws in a substantial excluded family rather than merely restating the exclusion.

The numerical calibration is now a finite algebraic problem with certified convex upper bounds and quantitative frame stability. This does not make spectral-gap positivity decidable. The general dominant-nongapped problem beyond the dual badly approximable planar families, sharp finite integer widths, uniform varying-signal laws and optimal table or fair-bit costs remain distinct

questions. The optional arithmetic LPS application in the preserved manuscript retains its original explicitly stated external dependency and its original-page audit limitation. Neither these results nor their reproducible builds assert an independent priority certification or an editorial outcome.

APPENDIX A. THE SMALLEST ORBIT CONTROLS UNIFORM CONCENTRATION

In this section G is a compact connected Lie group, $U : G \rightarrow O(V)$, and $V^G = \{0\}$. Put

$$(A.1) \quad s(U) = \min_{u \in \mathbb{S}(V)} \dim(Gu).$$

The minimum is over *all* directions. It can be smaller than the dimension of a generic orbit. Since G is connected, $s(U) \geq 1$.

Theorem A.1 (Uniform concentration across orbit types). *There is a finite constant $A_U \geq 1$ such that*

$$(A.2) \quad \sup_{u, v \in \mathbb{S}(V)} m_G\{g : \|U_g u - v\| \leq r\} \leq A_U r^{s(U)} \quad (0 < r \leq 1).$$

This exponent is optimal among uniform power upper bounds: for a unit u_0 with $\dim(Gu_0) = s(U)$ there are $c, r_0 > 0$ such that

$$(A.3) \quad m_G\{g : \|U_g u_0 - u_0\| \leq r\} \geq c r^{s(U)} \quad (0 < r < r_0).$$

No regular-orbit, constant-stabilizer, or simple-spectrum hypothesis is needed in (A.2).

Proof. Write $m = \dim G$, $s = s(U)$ and let $\mathfrak{g}$ be the Lie algebra. For $u \in \mathbb{S}(V)$ the differential

$$A_u : \mathfrak{g} \longrightarrow V, \quad X \longmapsto dU(X)u$$

has rank $\dim(Gu) \geq s$. Its s th singular value is continuous in u and is strictly positive on the compact sphere. Its minimum is therefore positive. We give a chart argument that also records why variation of the stabilizer causes no loss in the exponent.

Fix $(g_0, u_0) \in G \times \mathbb{S}(V)$. In a smooth coordinate chart $g = \psi(z, w)$, where $z \in \mathbb{R}^s$ and $w \in \mathbb{R}^{m-s}$, select s coordinate directions and a linear projection $L : V \rightarrow \mathbb{R}^s$ for which

$$D_z[LU_{\psi(z, w)}u]_{(g_0, u_0)} = M$$

is invertible. This is possible by the rank bound. By shrinking to a product of bounded open rectangles $Z \times W$ and a neighborhood J of u_0 , arrange

$$\|D_z[LU_{\psi(z, w)}u] - M\|_{\text{op}} \leq \frac{1}{2}\sigma_{\min}(M) \quad (z, w, u) \in Z \times W \times J.$$

Take Z convex. For fixed (w, u) , integration along the segment from z to z' gives

$$(A.4) \quad \|LU_{\psi(z, w)}u - LU_{\psi(z', w)}u\| \geq \frac{1}{2}\sigma_{\min}(M)\|z - z'\|.$$

If both points lie in the inverse image of the radius- r ball around v , their projected distance is at most $2\|L\|r$. This inverse image, on the fixed w slice, consequently has diameter at most $4\|L\|r/\sigma_{\min}(M)$, and hence s -dimensional Lebesgue measure at most a constant times r^s . Haar measure has a bounded smooth density in the chosen relatively compact chart. Integration in the remaining $m - s$ coordinates yields the same bound there.

Choose finitely many of these product neighborhoods covering $G \times \mathbb{S}(V)$. For each fixed u , those whose J contains u cover G ; summing the preceding bounds proves (A.2). All constants are independent of the center v . An empty inverse image contributes zero. This use of a finite cover, rather than a separate density estimate on every orbit, is what controls neighborhoods of singular orbit types.

For the converse, $Gu_0 \simeq G/G_{u_0}$ is a compact smooth embedded s -dimensional submanifold. The pushforward of Haar probability is its normalized invariant Riemannian volume: the induced metric is G -invariant and the invariant probability on this homogeneous space is unique. A smooth

local graph at u_0 has volume comparable to r^s in a small ambient ball. This proves (A.3). If a uniform estimate with exponent $s' > s$ held, comparison at u_0 as $r \downarrow 0$ would contradict it. $\square$

The rank computation in (A.1) is intrinsic to the representation. For example, given algebraic skew matrices $B_1, \dots, B_m$ spanning its infinitesimal action, the condition $s(U) \leq r$ is the existence of a real vector u with $\sum u_i^2 = 1$ and all $(r+1)$ -minors of $[B_1 u \cdots B_m u]$ zero. Thus it is an explicit real-algebraic decision problem for fully specified algebraic data; quantifier elimination applies [8]. This is not a test for a spectral gap, and no efficient algorithm in a succinct representation is asserted.

Corollary A.2 (Intrinsic occupation exponent). *Suppose the representation satisfies (B.2). In the setting of Lemma 2.2, $q_N \geq \kappa > 0$ implies*

$$(A.5) \quad \sum_{\substack{0 \leq t \leq N \\ \bar{K}_t \leq k}} g_t \leq 2048 D (16 A_U)^{2/s(U)} \kappa^{-3-2/s(U)} k^{2/s(U)}.$$

The estimate allows arbitrary hidden continuations and imposes no bound on the other cuts.

Proof. Apply Theorem B.5 with the uniform estimate (A.2). In particular the conditional means may change orbit type at every step; their directions still satisfy the same estimate. $\square$

APPENDIX B. AN ORBITAL ENTROPY OCCUPATION THEOREM

Lebesgue measure is used on $V = \mathbb{R}^D$ and normalized Haar measure on G . For a function on V , set

$$(B.1) \quad \mathcal{U}_a f(z) = f(U_a^{-1} z), \quad P_t = \sum_a p_t(a) \mathcal{U}_a, \quad \Pi f(z) = \int_G f(U_g^{-1} z) dm_G(g).$$

Thus $\mathcal{U}_a^* = \mathcal{U}_{a^{-1}}$, Π is an orthogonal projection, and $P_t \Pi = \Pi P_t = \Pi$. The probability law p_t need not be symmetric. Its required squared norm gap is

$$(B.2) \quad \|P_t f\|_2^2 \leq \|\Pi f\|_2^2 + (1 - g_t) \|f - \Pi f\|_2^2, \quad 0 \leq g_t \leq 1.$$

Invariance in this formula means G -invariance, not necessarily radial symmetry. In a reducible representation invariant functions can carry many parameters.

Lemma B.1 (Transfer of a group norm gap). *If right convolution $R_{p_t} F(g) = \sum_a p_t(a) F(ga)$ on $L^2(G)$ has mean-zero norm at most $\sqrt{1 - g_t}$, then (B.2) holds in every orthogonal representation on V .*

Proof. For each z apply the group inequality to $F_z(g) = f(U_g^{-1} z)$. Its mean is $\Pi f(z)$ and $R_{p_t} F_z(g) = P_t f(U_g^{-1} z)$. Integrate the resulting inequality in z . Orthogonal changes of variables and Fubini identify the three integrals with those in (B.2). First take bounded compactly supported f and then use density. This proof does not replace the full group gap by a gap in the defining representation. $\square$

B.1. The conditional-mean cost. Let G_D be a standard independent real Gaussian vector and let γ_τ be the density of τG_D .

Lemma B.2 (A one-sided Gaussian coupling inequality). *For bounded X and a finite-valued variable J , put $M = \mathbb{E}[X \mid J]$. For every $\tau > 0$,*

$$(B.3) \quad h(X + \tau G_D) - h(M + \tau G_D) \leq \frac{\mathbb{E}\|X - M\|^2}{2\tau^2}.$$

Proof. Write $f = \text{Law}(M) * \gamma_\tau$. Differentiation of the Gaussian mixture and the conditional covariance identity give

$$(B.4) \quad \nabla^2 \log f(y) = -\tau^{-2}I + \tau^{-4} \text{Cov}(M \mid M + \tau G_D = y) \succeq -\tau^{-2}I.$$

Hence $\phi(x) = -\mathbb{E} \log f(x + \tau G_D)$ has Hessian at most $\tau^{-2}I$. Taylor's inequality at M yields $\phi(X) \leq \phi(M) + \langle \nabla \phi(M), X - M \rangle + \|X - M\|^2 / (2\tau^2)$. The conditional expectation of the linear term given J is zero. If $g = \text{Law}(X) * \gamma_\tau$, nonnegativity of $D(g \parallel f)$ gives $h(g) \leq -\int g \log f = \mathbb{E} \phi(X)$, whereas $\mathbb{E} \phi(M) = h(f)$. Boundedness of X, M provides Gaussian bounds with finite second moments and justifies differentiation and integration. These identities prove (B.3). $\square$

This elementary consequence of Gaussian posterior identities is used as an estimate, not as a claim of priority over the Gaussian estimation literature [1, 4]. The estimate is one-sided and can have a negative left side. Here J can be a finite register with no Gaussian observation model; the smoothing noise is added after conditioning.

Lemma B.3 (Entropy production away from invariant functions). *For any bounded-support Gaussian mixture density f ,*

$$(B.5) \quad h(P_t f) - h(f) \geq g_t \|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq 0.$$

Proof. Put $f_a = \mathcal{U}_a f$ and $h = P_t f$. Orthogonality preserves entropy, so $h(h) - h(f) = \sum_a p_t(a) D(f_a \parallel h)$. For probability densities u, v , $D(u \parallel v) \geq \|\sqrt{u} - \sqrt{v}\|_2^2$; it follows, for example, from $-\log x \geq 2(1 - \sqrt{x})$ and integration. The mean is the least-squares minimizer, and thus

$$\sum_a p_t(a) \|\sqrt{f_a} - \sqrt{h}\|_2^2 \geq \sum_a p_t(a) \|\sqrt{f_a} - P_t \sqrt{f}\|_2^2 = 1 - \|P_t \sqrt{f}\|_2^2.$$

Apply (B.2) and $\|\sqrt{f}\|_2 = 1$. $\square$

No logarithmic Sobolev inequality or point-mass mixing assertion is used in this step.

B.2. Sparsity away from invariant densities. Suppose that the representation obeys the uniform orbital bound

$$(B.6) \quad m_G\{g : \|U_g u - v\| \leq r\} \leq A r^s \quad (u, v \in \mathbb{S}(V), \ 0 < r \leq 1),$$

where $A \geq 1$ and $s > 0$. In particular the representation has no nonzero fixed vector. It is essential that u range over all possible centroid directions, not merely the initial seed orbit.

Lemma B.4 (Finite centroids cannot be almost invariant). *Suppose Z has at most k values in the unit ball and $\mathbb{E}\|Z\| \geq \kappa$, where $0 < \kappa \leq 1$. Set*

$$(B.7) \quad r = \left(\frac{\kappa}{16Ak} \right)^{1/s}, \quad \tau = \frac{\kappa r}{8\sqrt{2D}}, \quad f = \text{Law}(Z) * \gamma_\tau.$$

Then $\|\sqrt{f} - \Pi \sqrt{f}\|_2^2 \geq \kappa/16$.

Proof. Atoms of norm at least $\kappa/2$ have total mass at least $\kappa/2$. Let Ω be the union of radius- r caps around their directions, and let C_Ω be its cone. By Markov's inequality, $\mathbb{P}(\|G_D\| \leq \sqrt{2D}) \geq 1/2$. On this event, perturbing a high-norm atom by τG_D changes its direction by at most $4\tau\sqrt{2D}/\kappa = r/2$. Therefore $\int_{C_\Omega} f \geq \kappa/4$.

For any G -invariant $u \in L^2(V)$, Haar averaging and (B.6) give

$$\int_{C_\Omega} |u(z)|^2 dz = \int |u(z)|^2 \int_G \mathbf{1}_{C_\Omega}(U_g z) dm_G(g) dz \leq k A r^s \|u\|_2^2.$$

The point $z = 0$ is Lebesgue-null. Taking $u = \Pi \sqrt{f}$, whose norm is at most one, bounds this integral by $\kappa/16$. The reverse triangle inequality in $L^2(C_\Omega)$ now gives a norm difference at least $\sqrt{\kappa/4} - \sqrt{\kappa/16} = \sqrt{\kappa}/4$. Repeated directions and zero centroids cause no change. $\square$

Theorem B.5 (Orbital occupation budget). *Under (B.2) and (B.6), let a finite hidden register track a unit-vector command process as in (2.1). If $q_N \geq \kappa > 0$, then, for every $k \geq 1$,*

$$(B.8) \quad \sum_{\substack{0 \leq t < N \\ K_t \leq k}} g_t \leq B_{D,A,s} \kappa^{-3-2/s} k^{2/s}, \quad B_{D,A,s} = 2048D(16A)^{2/s}.$$

The same inequality holds with K_t replaced by the number of positive-probability labels under the chosen test law. No bound is imposed on the other epochs.

Proof. Fix k and the single scale (B.7) for the entire run. Write $f_t = \text{Law}(Z_t) * \gamma_\tau$. Isotropy of the Gaussian makes the density of $U_a Z_t + \tau G_D$ equal to $P_t f_t$. By Lemmas B.2 and B.3,

$$h(f_{t+1}) - h(f_t) \geq g_t \|\sqrt{f_t} - \Pi \sqrt{f_t}\|_2^2 - \frac{\Delta_t}{2\tau^2}.$$

At an epoch with at most k labels, Lemma B.4 bounds the norm term below by $\kappa/16$. Elsewhere it is nonnegative. The initial density is a translate of γ_τ ; the terminal second moment is at most $1 + D\tau^2$. The Gaussian maximum-entropy bound and (2.2) yield

$$\frac{\kappa}{16} \sum_{K_t \leq k} g_t \leq \frac{D}{2} \log \left(1 + \frac{1}{D\tau^2} \right) + \frac{1}{2\tau^2} \leq \frac{1}{\tau^2}.$$

Substitute (B.7). The same Gaussian scale is retained through all epochs; no repeated mixing-time charge is incurred. $\square$

Corollary B.6 (Actual terminal error). *For a constant gap $g > 0$, let $\mathcal{E}_N(\mathbf{K})$ be the minimum actual wordwise binary-TV error over a prescribed available profile, and put $K = \max_t K_t$. Then*

$$(B.9) \quad \mathcal{E}_N(\mathbf{K}) \geq \frac{1}{2} \left[\frac{\rho}{\sqrt{D}} - \left(\frac{B_{D,A,s} K^{2/s}}{gN} \right)^{1/(3+2/s)} \right]_+.$$

In particular, $K = o(N^{s/2})$ implies $\liminf \mathcal{E}_N \geq \rho/(2\sqrt{D})$.

Proof. If the actual error is below the calibration threshold, combine (2.3) with (B.8) and solve for it. Otherwise the displayed lower bound already holds. Attainment or a minimizing sequence gives the same conclusion for the optimum. $\square$

Local simulation errors may cancel. Nothing in this corollary sums such errors or rules out cancellation; its objective is the actual final law.

APPENDIX C. EXACT SYNTHESIS FROM A MINIMAL ORBIT

Assume in this section that V is a nontrivial real irreducible orthogonal representation, of dimension d . Choose a unit vector v_0 with orbit dimension $s = s(U)$, and set

$$\mathcal{O} = Gv_0, \quad C = \text{conv } \mathcal{O}.$$

The orbit may be a singular orbit in the unit sphere. As a homogeneous space in its own right it is nevertheless a smooth compact manifold.

Lemma C.1 (Inradius and quadratic support approximation). *One has $d^{-1}\mathbb{B}(V) \subseteq C \subseteq \mathbb{B}(V)$. There are $c, C_0, h_0 > 0$, depending on U and v_0 , such that for $0 < h \leq h_0$ a set $\mathcal{V}_h \subset \mathcal{O}$ with $|\mathcal{V}_h| \leq C_0 h^{-s}$ satisfies*

$$(C.1) \quad (1 - ch^2)C \subseteq \text{conv } \mathcal{V}_h \subseteq C \quad \text{whenever } ch^2 < 1.$$

Proof. The mean of $U_g v_0$ is an invariant vector and is zero. Its covariance is I/d : the covariance is self-adjoint and commutes with the representation, so its eigenspaces are invariant; irreducibility and trace one give the claim. This is the real Schur orthogonality calculation, for example [7, Lemma 5]. For unit H , let $f(g) = \langle H, U_g v_0 \rangle$ and $M = \max_g f(g)$. Since $-1 \leq f \leq M$,

$$0 \leq \mathbb{E}[(M - f)(f + 1)] = M - \mathbb{E}f^2 = M - 1/d.$$

Every support function of C is thus at least $1/d$, which proves the inradius statement.

Equip $\mathcal{O}$ with its induced Riemannian metric. Finitely many bounded smooth coordinate patches, with their inverses uniformly Lipschitz on smaller patches, give an intrinsic h -net with at most $C_0 h^{-s}$ points. At a maximizer y of $z \mapsto \langle H, z \rangle$ on $\mathcal{O}$, its tangential first derivative is zero. The second fundamental form of the compact embedded orbit is bounded. Along a unit-speed minimizing geodesic of length at most h , Taylor's formula therefore gives

$$\langle H, y \rangle - \langle H, v \rangle \leq L \|H\| h^2$$

for a nearby net point v , with L independent of H, y . The inradius bound gives $\|H\| \leq d h_C(H)$. Hence $h_{\text{conv } \mathcal{V}_h}(H) \geq (1 - L d h^2) h_C(H)$ for every H . Taking $c \geq L d$, and using the support-function criterion, proves (C.1). $\square$

The covering exponent s and quadratic loss at a support maximum are geometric facts, not new quantization exponents. What is needed for realization is an enclosure compatible with *every command* and *every layer*; the following construction supplies it.

Theorem C.2 (A minimal-orbit positive realization). *For any finite command alphabet, any finite unit seed set, and $0 < \rho \leq 1/(8d)$, the experiment (1.1) has an exact realization with at most $C_U N^{s(U)/2}$ available labels, for $N \geq 1$. All command matrices can be chosen independent of the epoch; the final decoder may depend on N . Each row has at most $d + 1$ successors. If the seeds belong to $C \cap \mathbb{S}(V)$, the amplitude range improves to $0 < \rho \leq 1/10$.*

Proof. Choose

$$h_N = \min\{h_0, (8cN)^{-1/2}\}, \quad \eta = 1 - c h_N^2 \geq 1 - 1/(8N), \quad a_t = \eta^{-t}/4.$$

Let $v_1, \dots, v_M$ be the net points. The represented vector of a label i at layer t is $a_t v_i$. These vectors describe the fixed rows; they are not stored as real-valued runtime data.

The polytope $\text{conv}\{v_i\}$ contains ηC and hence a ball of radius η/d . For a unit seed x , $4\rho x \in \eta C$ if $\rho \leq 1/(8d)$, because $4\rho \leq 1/(2d) < \eta/d$. Choose its convex coordinates as the initial distribution. More generally, if $x \in C$ and $4\rho \leq 2/5 < \eta$, the same initialization is valid. For each command a and label i , invariance of C gives

$$\eta U_a v_i \in \eta C \subseteq \text{conv}\{v_j\}.$$

Choose any convex coordinates of this point for the row $T_a(i, \cdot)$. They do not depend on t . Since $a_{t+1}\eta = a_t$, conditional expectation updates the represented vector by exactly U_a . Induction gives mean $\rho U_w x$ after each complete prefix.

Use the terminal column $d_j(i) = a_N \langle e_j, v_i \rangle$. Bernoulli's inequality gives $\eta^N \geq 7/8$, so $|d_j(i)| \leq 2/7 < 1$. This proves every numerical conditional law exactly. Carathéodory's theorem gives $d + 1$ successors. Finally $M \leq C_0 h_N^{-s} \leq C_U N^{s/2}$, with the fixed two-state output cut absorbed into the constant. Neither convex feasibility nor row lookup is priced as computation in this model. $\square$

Corollary C.3 (The irreducible width law). *Suppose a probability law on the finite command alphabet has a full mean-zero group norm gap. For a nontrivial real irreducible V , any nonempty finite unit seed set, fixed $0 < \rho \leq 1/(8d)$, and $0 \leq \varepsilon < \rho/(2\sqrt{d})$,*

$$(C.2) \quad W_{N,\varepsilon} = \Theta_{U,p,\rho,\varepsilon}(N^{s(U)/2}).$$

The same order holds for exact vector-state realizations and for fixed-length realizations with common command rows. For seeds on a minimal orbit the amplitude range is $0 < \rho \leq 1/10$.

Proof. A full group norm gap transfers to (B.2) with a fixed $g > 0$. Apply Corollary A.2 at the peak k , using $\kappa = \rho/\sqrt{d} - 2\varepsilon > 0$. It gives $gN \leq B\kappa^{-3-2/s}k^{2/s}$ and therefore the lower order. The exact machine in Theorem C.2 is an upper bound for every allowed error, is vector-valued in its representation, and has common command rows. These subclasses lie inside the unrestricted clocked class, so the lower bound applies to them too. The last assertion uses the stronger initialization in that theorem. $\square$

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